Jiva: Increasing profitability for millions of smallholder farmers

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By leveraging Go
Jiva helps smallh
farmers access a
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About Jiva

Jiva was founded in 2020 as a technology-enabled platform and mobile app to improve the livelihoods of the 525 million smallholder farmers who grow 70% of the region's food. Jiva offers free personalized agronomic advice, extends cash advances for farm inputs, sells and delivers high-quality inputs, and purchases crops at harvest. It is backed by Olam International, a global food and agri-business company.

Industries: Agriculture, Forestry & Fishing

Google Cloud results

- Reduces information asymmetry for millions of Indian and Indonesian smallholder farmers
- Increases crop yield by up to 50%
- Lowers agricultural inputs required by up to 20%
- Speeds up deployment to production by reducing code commit time to 15 minutes

Improv smallh farmer profita 25%

Location: Singapore

Run (https://cloud.google.com/run/),

(https://cloud.google.com/kubernetes-engine/),

Vertex AI (https://cloud.google.com/vertex-ai/),

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- Google Cloud (https://cloud.google.com/)
- Cloud Bigtable (https://cloud.google.com/bigtable/)
- Cloud Run (https://cloud.google.com/run/)
- Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)
- TensorFlow (https://www.tensorflow.org/)
- Vertex AI (https://cloud.google.com/vertex-ai/)
- Cloud SQL (https://cloud.google.com/sql/)
- Cloud Storage (https://cloud.google.com/storage/)
- Firebase (https://firebase.google.com/)
- Compute Engine (https://cloud.google.com/compute/)

In our modern world, farming is becoming less visible as fewer people take to agriculture as a career. According to the World Bank (https://data.worldbank.org/indicator/SL.AGR.EMPL.ZS?name_desc=false), employment in agriculture as a percentage of total employment has dropped from 44% to 27% globally in the past three decades.

According to Seamus Tardif, Head of Growth and founding team at Jiva (https://www.jiva.ag/), one of the reasons for this challenge is due to the lack of profitability in the industry.

"If you can't make farming more profitable for farmers, they will continue to flee from farming and start moving into more urban-based jobs, which creates a massive food security issue," he explains.

"Rural farmers have been affected by information asymmetry. Local collectors could manipulate prices because they can leverage the information disparity between the rural and urban environments. As a result, farmers were always in a buyer's market and never a seller's market," adds Tardif.

Jiva currently operates in two markets, Indonesia and India. The former is an archipelagic country, and the latter comprises 28 states, each with different government administrations.

To handle the deep complexity brought about by these large geographies, Jiva integrated with Google Cloud (https://cloud.google.com/) in 2020, and it has been progressively adopting new tools to manage the challenging Indonesian and Indian agricultural landscapes.

Serving farmers where they are

"Large geographic and climate diversity makes Jiva a very customized business, because farms that are merely 60 miles apart may need to meet different volumes of agricultural inputs or means of calculating crop prices," explains Tejas Dinkar, Chief Technology Officer of Jiva.

Furthermore, Jiva aims to help farmers in four key areas, which include access to adequate financing, high-quality agricultural inputs to improve yields, customized agronomic advisory services, and a platform for farmers to sell their crops themselves.

"Working inside rural environments

makes cloud-based infrastructure super important, especially since many of these countries have internet connections with low bandwidth."

—Seamus Tardif, Head of Growth and Founding Team, Jiva

It would have been prohibitively slow and expensive to serve farmers in these rural locations without Google Cloud infrastructure to enable Jiva's different systems to scale products and services that need to call back to central databases.

"Working inside rural environments makes cloud-based infrastructure super important, especially since many of these countries have internet connections with low bandwidth," explains Tardif.

Handling complex agricultural needs with a lean team

Despite having a lean team, Jiva ensures they deliver the right information to farmers, no matter where they are. To do so, they require the flexibility to create a unique service mesh, so they can plug in different bits of logic that apply to different geographies.

"With Google Cloud, we realized we can do much more with fewer engineers. For the longest time, we didn't even have a full-time DevOps person on staff. We had one DevOps person working half-day. And, even with that, our part-time DevOps engineer could set up and run seven different microservices for the different verticals at Jiva," Dinkar shares.

"GKE has allowed me to cascade responsibility back down to individuals rather than requiring a very heavy DevOps team. As a result, I can focus on developing new, more complex products that we will use as we expand our geographic footprint."

—Tejas Dinkar, Chief Technology Officer, Jiva

Jiva is heavily affected by crop seasonality and must ensure flexible service delivery at the start of a season after farmers have not used their app for several months. To accomplish this, Jiva relies heavily on Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE) to manage the team's entire Kubernetes infrastructure.

GKE enables Jiva's developers to take ownership of their DevOps practice through Helm Charts that are checked into a repository with the code base within a single artifact. As a result, each developer can easily increase scalability, manage alerts, and tweak configurations.

"GKE has allowed me to cascade responsibility back down to individuals rather than requiring a very heavy DevOps team. As a result, I can focus on developing new, more complex products that we will use as we expand our geographic footprint," says Dinkar.

Jiva's engineers have reduced the time from code commit to deployment in production to 15 minutes, which is particularly helpful when field personnel provide feedback that an element of their service network, or a piece of logic, is not properly implemented and a quick fix is required.

Furthermore, Jiva's engineers can also rely on Cloud SQL (https://cloud.google.com/sql) and Cloud Storage (https://cloud.google.com/storage/docs/caching) for their database needs because it is fully managed, meaning these engineers do not need to worry about backups and uptime.

The Jiva app delivers a Crop Doctor service built with Vertex AI (https://cloud.google.com/vertex-ai) that leverages computer vision and image processing to determine crop diseases and their causes, like potassium deficiency, and suggests remedies that traditional farmers would not be aware of.

"This is also geography dependent as Indonesia is largely tropical, while India is both tropical and temperate, so the solution to the same diseases or even the diseases in some cases might be very different," explains Arnav Pandey, Chief Product Officer of Jiva.

Building a smarter app for the future of agriculture

Currently, Jiva is trialing the use of TensorFlow to develop better credit-worthiness and pricing models for farmers.

"In rural areas, scoring credit-worthiness is hampered by a lack of clear or enough data around credit history. But we can also assess other indicators like purchasing patterns and family debt history. This enables us to better credit profile our stakeholders," Pandey says.

He further explains that Jiva's revamped pricing model seeks to prioritize price competitiveness by factoring in variables like defaults, retailer or micro collector fulfillment, and SKU prices.

Even as the team builds on its integration with Google Cloud moving forward, it is already helping farmers to bear much fruit. Its recent farm surveys indicate that farmers are earning up to 25% more annual income, increasing their yield by up to 50%, and reducing the amount spent on inputs by nearly 20%.

"With Google Cloud, we can cover more territory without scaling our team massively. And what that means is that we're able to spend money in key places where it's necessary for us to be able to serve more farmers and ensure that profitability is going directly back to farmers."

—Seamus Tardif, Head of Growth and Founding Team, Jiva

Jiva is looking to expand its geographic footprint with its four-pronged approach to transform farmers' financial security, and turn farming into a profitable and empowering path for rural folk.

"With Google Cloud, we're able to cover more territory without the risk of scaling the team massively. And what that means is that we're able to spend money in key places where it's necessary for us to be able to serve more farmers and ensure that profitability is going directly back to farmers," Tardif concludes.